Demystifying Blockchains

A blockchain is a continuously growing list of records, called blocks, which are linked and secured using cryptography to ensure data security.

Data security is of paramount importance to corporations. Enterprises need to establish high levels of trust and offer guarantees on the security of the data being shared with them while interacting with other enterprises. The major concern of any enterprise about data security is data integrity. What many in the enterprise domain worry about is, "Is my data accurate?"

Data integrity ensures that the data is accurate, untampered with and consistent across the life cycle of any transaction. Enterprises share data like invoices, orders, etc. The integrity of this data is the pillar on which their businesses are built.

Blockchain

A blockchain is a distributed public ledger of transactions that no person or company owns or controls. Instead, every user can access the entire blockchain, and every transaction from any account to any other account, as it is recorded in a secure and verifiable form using algorithms of cryptography. In short, a blockchain ensures data integrity.

A blockchain provides data integrity due to its unique and significant features. Some of these are listed below.

***Timeless validation for a transaction*:** Each transaction in a blockchain has a signature digest attached to it which depends on all the previous transactions, without the expiration date. Due to this, each transaction can be validated at any point in time by anyone without the risk of the data being altered or tampered with.

***Highly scalable and portable*:** A blockchain is a decentralised ledger distributed across the globe, and it ensures very high availability and resilience against disaster.

***Tamper-proof*:** A blockchain uses asymmetric or elliptic curve cryptography under the hood. Besides, each transaction gets added to the blockchain only after validation, and each transaction also depends on the previous transaction. Hence, a blockchain is nearly impossible to tamper with without anyone noticing.

Demystifying blockchains

A blockchain, in itself, is a distributed ledger and an interconnected chain of individual blocks of data, where each block can be a transaction, or a group of transactions.

In order to explain the concepts of the blockchain, let's look at a code example in JavaScript. The link to the GitHub repository can be found at *https://github.com/abhiit89/AngCoins*. So do check the GitHub repo and go through the 'README' as it contains the instructions on how to run the code locally.

***Block*:** A block in a blockchain is a combination of the transaction data along with the hash of the previous block. For example:

```
class Block {
    constructor(blockId, dateTimeStamp, transactionData,
previousTransactionHash) {
        this.blockId = blockId;
        this.dateTimeStamp = dateTimeStamp;
        this.transactionData = transactionData;
        this.previousTransactionHash =
previousTransactionHash;
        this.currentTransactionHash = this.
calculateBlockDigest();
    }
```

The definition of the block, inside a blockchain, is presented in the above example. It consists of the data (which includes *blockId*, *dateTimeStamp*, *transactionData*, *previousTransactionHash*, *nonce*), the hash of the data (*currentTransactionHash*) and the hash of the previous transaction data.

Genesis block: A genesis block is the first block to be created at the beginning of the blockchain. For example:

```
new Block(0, new Date().getTime().valueOf(), 'First Block',
'0');
```

Adding a block to the blockchain

In order to add blocks or transactions to the blockchain, we have to create a new block with a set of transactions, and add it to the blockchain as explained in the code example below:

```
addNewTransactionBlockToTransactionChain(currentBlock) {
        currentBlock.previousTransactionHash = this.
```